

Thamidu Nadun

Full-Stack Developer

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LinkedIn GitHub Blog

About Me

Self-taught full-stack developer and undergraduate technologist with strong expertise in React, Spring Boot, and Node.js. Skilled in backend development, REST APIs, database integration, and IoT systems with ESP32. Experienced in AI/ML model development, including practical applications such as recommendation systems. Continuously expanding knowledge in Three.js, WebXR, Rust, cybersecurity, and cloud technologies, with a focus on building functional and reliable software solutions.

Skills

- React.js
- Three.js
- WebAR
- Rest API Development
- MySQL
- Machine Learning
- Desktop App development
- Spring Boot
- IoT Development
- AWS, GCP
- Team Management
- Problem Solving

languages

English

Sinhala

Experience

Full-Stack Developer — (Freelance)

Apr 2025

YASMEEN (PVT) LTD

- Developed a bilingual (EN/JP) invoice automation platform using React, Flask, and Playwright, **replacing a manual Excel-based workflow** for a Japan-focused tourism company.
- Reduced PDF generation latency from 2–3 minutes to 6–7 seconds (~95%) by implementing persistent Chromium rendering and eliminating browser startup overhead.
- Automated a **1 hour invoice processing workflow into a 3 minute email-driven pipeline**, including data extraction, invoice generation, PDF rendering, and email delivery.
- Built language-aware email parsers for English and Japanese correspondence, enabling automated invoice creation from unstructured multilingual emails.
- Designed reusable headless rendering services and workflow automation components that improved scalability, reliability, and operational efficiency.
- Built an **AI powered translation and data normalization pipeline** that *converted Japanese itinerary information* into structured JSON, enabling automatic generation of driver itinerary PDFs from email content.

Projects

Personal Blog

Nov 2025 - Jan 2026

Full-Stack Project

- Engineered a **full-stack blogging platform with a RESTful backend and SPA frontend**, supporting posts, tags, and user management.
- Implemented **JWT-based authentication and role-based access control (RBAC)** for admin, author, and reader workflows.
- Designed normalized database schemas and optimized API queries, enabling fast content retrieval and smooth UI rendering.

Movie Recommendation System

Sep 2025 - Oct 2025

Machine Learning

- Built an **unsupervised recommendation pipeline** using *K-Means, Hierarchical Clustering, and Gaussian Mixture Models*
- Performed **data preprocessing**, **feature engineering**, and cluster evaluation to improve recommendation relevance.
- Compared multiple models and selected optimal configurations based on silhouette score

Greenhouse Monitoring System

Dec 2024 - Jan 2025

IoT

- Developed an end-to-end IoT monitoring and control system using **ESP32 and React.js**
- Collected and streamed environmental data (temperature, humidity, soil moisture) to a live web dashboard.
- Allowed remote actuation of pumps, fans, integrating embedded devices with web technologies.

Notedeck – Markdown Presentation — Oct 2025 - Jan 2026

SaaS

- Developed a SaaS tool that converts Markdown into interactive HTML presentations.
- Enabled importing content directly from GitHub URLs, automating presentation creation.
- Focused on developer productivity by simplifying slide authoring workflows.

Malware Development/AV Evasion — Feb 2026

cyberSecurity

- Developed a Windows-based keylogger in C as part of malware development and security research activities.
- Implemented keystroke capture, logging, and persistence mechanisms using native Windows APIs.
- Explored **static detection evasion techniques such as string obfuscation and signature reduction** to study antivirus detection methods.
- Analyzed how security products identify and respond to malicious behavior through controlled testing and experimentation.

tinyWebC – HTTP Server from Scratch — Mar 2026

Low-Level System

- Engineered a **web server in C** from first principles, handling TCP connections, HTTP request parsing, and response generation.
- Implemented **static file serving with MIME type resolution**, enabling browsers to correctly render HTML, CSS, JavaScript, and image resources.
- Developed core server components including socket management, file I/O, string processing, and dynamic response construction.
- Applied systems programming principles such as memory management, network communication, and protocol implementation to build a functional web server with minimal dependencies.

Whatsapp Bot — Apr 2025 - Apr 2025

Automation

- Designed a command-driven automation platform supporting text, media, and external knowledge services.
- Applied modular slash commands enabling users to retrieve web images, generate AI images, fetch scientific data, and access dictionary definitions.
- Created reusable command routing and execution pipelines, making new features easy to plug in.

Presentee - Presentation Highlighter App — Apr 2025

SaaS

- Developed Presentee, a cross-platform desktop app that enhances live presentations with stylish, gradient-highlighted sections, targeting presenters who want visually engaging presentation.
- Built with Electron, HTML, CSS, and JavaScript, enabling smooth overlays, animated borders, and interactive presentation enhancements.

Education

BSc (Hons) in Information Technology Specialising in Cyber Security — 2024 - 2027

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